

MAT 310, Handout 12: Cumulative Distribution Functions, Continuous RVs (4.1-4.3)

Cumulative Distribution Function (cdf) The cdf of any RV Y is given by

$$F(y) = P(Y \leq y)$$

for $-\infty < y < \infty$.

Example 1. Find the cdf of a random variable Y whose distribution is given by:

y	$p(y)$
1	.2
2	.5
3	.3

Remark: The cdf $F(y)$ of a RV Y has jump discontinuities at values y such that $p(y) > 0$. Therefore, the cdf of a discrete RV is a _____. That is, $F(y)$ increases at only a finite (or, perhaps, countably infinite) number of points.

Properties of the cdf

1. $\lim_{y \rightarrow -\infty} F(y) =$
2. $\lim_{y \rightarrow \infty} F(y) =$
3. $F(y)$ is _____
4. $F(y)$ is _____-continuous

One more property: To be mathematically precise, the first derivative of $F(y)$ needs to exist and be continuous except for, at most, a finite number of points in any finite interval. All the distribution functions for the continuous RVs discussed in the textbook satisfy this requirement.

Definition: A RV is **continuous** if its cdf $F(y)$ is continuous for all y .

Remark: Unlike discrete RVs, a continuous RV can take on any value in an interval.

Note: If Y is a continuous RV, then $P(Y = y) = 0$ for all y .

Probability Density Function (pdf) The pdf of a continuous RV Y is given by

$$f(y) = \frac{dF(y)}{dy} = F'(y).$$

Properties of the pdf

1. $f(y) \geq 0$ for all y (since $F(y)$ is nondecreasing and, therefore, $F'(y) \geq 0$)
2. $\int_{-\infty}^{\infty} f(y) dy = 1$

Question: Given $f(y)$, how to find $F(y)$?

Example 2. Let Y be a continuous RV with pdf defined by

$$f(y) = \begin{cases} y, & 0 < y < 1 \\ c - y, & 1 \leq y < 2 \\ 0, & \text{o.w.} \end{cases}.$$

(a) Find c .

(b) Find $F(y)$.

Question: How to compute $P(a \leq Y \leq b)$ for a continuous RV Y ?

Percentiles: Let ϕ_p be the $(100 \cdot p)^{th}$ percentile of RV Y . We define $\phi_p = \min\{y \mid F(y) \geq p\}$. So, $\phi_{0.5}$ is the median of RV Y .

Example 3. Consider the RV in Example 2.

(a) Find the median of Y .

(b) Find the 70th percentile of Y .

(c) Find $P(0.5 \leq Y \leq 1.5)$.

The **expected value** of a continuous RV Y is given by:

$$E(Y) =$$

The **expected value** of a function $g(Y)$ is given by:

$$E[g(Y)] =$$

The **variance** of a continuous RV Y is given by:

$$V(Y) =$$

Remark: The properties of expected values and variances for discrete random variables from Handout 6 also hold for continuous random variables (with analogous proofs).

Example 4. Find $E(Y)$ and $V(Y)$ for the RV in Example 2.